OO-M-340D
15 May 1986
SUPERSEDING
OO-M-340C
June 25, 1980

FEDERAL SPECIFICATION

MILLING, GRINDING, DRILLING, AND SLOTTING ATTACHMENT

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an electric motor driven attachment for milling, grinding, drilling, slotting, boring, gear cutting, thread milling, and keyseating operations. The attachment is adaptable for mounting on a lathe, planer, or boring mill. This attachment may also be used as a portable, self-contained unit suitable for bench mounting when provided with a feed table.

1.2 Classification.

1.2.1 Sizes. The attachment covered by this specification shall be of the following sizes, as specified (see 6.2):

Size 1 - See Table I.
Size 2 - See Table I.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3460

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Federal Standards

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-123	-	Marking for Shipment (Civil Agencies)
FED-STD-376	-	Preferred Metric Units for General Use by the Federal Government

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes, are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-T-152	-	Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	-	Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-E-16298	-	Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-M-18058	-	Machinery, Metal and Woodworking, Packaging of

Military Standards

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulation:

Occupational Health and Safety Administration (OSHA) 29 CFR 1910.211-222
General Industrial Safety and Health Standards

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(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Gear Manufacturers' Association (AGMA)

AGMA 360.02	-	Manual for Machine Tool Gearing
AGMA 390.03	-	Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Dr., Arlington, VA 22209.)

Amercian National Standards Institute (ANSI)

ANSI B5.10	-	Machine Tapers
ANSI B7.1	-	Safety Requirements For The Use, Care, and Protection of Abrasive Wheels
ANSI B17.1	-	Keys and Keyseats
ANSI B94.19	-	Milling Cutters and End Mills

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM)

ASTM D3951-82	-	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

International Organization for Standardization (ISO)

ISO 54	-	Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches
ISO 1122/1	-	Glossary of Gear Terms - Part 1: Geometrical Definitions

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Electrical Manufacturers' Association (NEMA)

NEMA ICS 1	-	General Standards for Industrial Control and Systems

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(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

National Fire Protection Association (NFPA)

NFPA 70	-	National Electrical Code
NFPA 79	-	Electrical Standard for Metalworking Machine Tools and Plastics Machinery

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

Rubber Manufacturers' Association (RMA)

RMA IP 25	-	Variable Speed Belts

(Application for copies should be addressed to the Rubber Manufacturers' Association, 1400 K Street, NW, Washington, DC, 20005.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one or more complete attachments for first article inspection (see 4.3 and 6.2). The first article may be either a standard commercial production model, preproduction model, or initial production item which conforms to the requirements of this specification. In any case, the approved first article and the production items shall be in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The attachments shall be new and of the manufacturer's latest design which shall meet the requirements of this specification. The attachments shall be powered and capable of multipositional, vertical to horizontal, spindle axis adjustments for machine operations and able to mount on a lathe, milling machine, planer, and boring mill. The attachments shall be portable and self contained when mounted upon a compound slide. When mounted on a lathe, the attachments shall be adaptable for mounting on the compound rest or cross slide. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair without modification, distortion, or damage to the attachment or any attachment part or component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the attachments. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in FED-STD-376.

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3.2.2 Measuring and indicating devices. When specified (see 6.2), measuring and indicating devices shall be graduated in both the US and SI systems of measurement. When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2). Regardless of the measurement system used, all measuring and indicating devices shall be graduated in the same system.

3.2.3 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.4 Gears. All gears and pinion designs selected for use in the attachment or its components shall be manufactured to either English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module system in accordance with AGMA 600.01.

3.2.5 Safety devices. Safety devices shall be provided for parts that present safety hazards to personnel. Safety devices shall not interfere with the operation of the attachment or with the operator's performance.

3.2.5.1 Occupational safety standards. The product shall comply with OSHA standards set forth in 29 CFR 1910.211-222 that are applicable to this type of equipment.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. Parts used in fabrication of the attachments shall be new and free of defects. Part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Screws, pins, bolts, and similar parts shall be installed with means to prevent loss of tightness and adjustment. Parts subject to adjustment or removal shall not be swaged, peened, staked, or otherwise deformed.

3.4.1 Castings and forgings. Castings shall be free from visible blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or

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other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Only rosin base fluxes shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.4.4 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.4). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the attachment to remove the temporary caps or stoppers before start-up or use of the attachment.

3.4.5 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.4.6 Instruction plates. Instruction plates shall be permanently marked with bold-faced English letters on contrasting backgrounds and shall conform to MIL-STD-130.

3.5 Performance and product characteristics. The basic attachments, with appropriate standard and optional equipment, shall be capable of performing plain and universal milling, drilling, and boring operations through vertically adjustable three axis spindle positioning and shall be capable of standard keyseating or slotting with the required tooling. Feed for cutting operations shall be obtained from the co-joined machine except as otherwise specified herein.

3.5.1 Capacities. The capacities of the attachment set shall not be less than the requirements specified in Table I.

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TABLE I. Capacities.

Requirement	Size 1	Size 2
Adaptable for use on lathes of (inch) swing	10 to 13	13 to 18
Mounting surface to spindle center (inches)	1 13/16	2 5/8
Vertical adjustment (inches)	5	6
Spindle taper (Morse, ANSI B5.10)	2	3
Range of speed, (rpm)	44 to 450	44 to 450
Optional speed* range (rpm)	540 to 5500	540 to 5500
Number of speeds (range)	13 (Minimum)	13 (Minimum)
Motors (see 3.6.4.1)	1/3 hp 115/230 volts 50/60 Hz Single phase	3/4 hp 115/230 volts 50/60 Hz Single phase
Optional motor*	-	3/4 hp 208/220/440 volts 50/60 Hz Three phase

*Optional speed range and/or optional motor shall be furnished when so specified (see 6.2).

3.5.2 Accuracy. The head shall be vertically adjustable and means shall be provided to lock the head at any point along the vertical travel without damaging the guidepost(s). The vertical adjustment shall work freely through the length of travel without sticking or binding. Locking shall be accomplished without deflecting the tool position more than 0.001 inch. The error in adjustment shall not exceed 0.0005 inch per inch of travel.

3.5.3 Milling with basic attachment. The basic attachment, when mounted on a suitable lathe, shall permit the following operations within the specified tolerances.

3.5.3.1 Woodruff keyseat cutting. The basic attachment shall be capable of cutting Woodruff keyseats of the sizes specified below in 1020 steel, round bar stock mounted between lathe centers, in not more than five minutes for each cut. One keyseat shall be cut 1/4 inch from the end of the bar to the keyseat center. A second keyseat shall be cut at the midpoint of the bar. The cuts shall be free of chatter marks.

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Size 1 - No. 204 Woodruff cutter, bar size 3/4 inch diameter by 6 inches long.

Size 2 - No. 406 Woodruff cutter, bar size 1 inch diameter by 8 inches long.

3.5.3.2 Keyway cutting. The basic attachment shall be capable of cutting keyways of the sizes specified below in one continuous operation in 1020 steel, round bar stock, 30 inches long, mounted between centers.

Size 1 - 1/2 x 1/4 x 25 inches long at 1 3/4 inches per minute (ipm), bar diameter two inches.

Size 2 - 3/4 x 3/8 x 25 inches long at 1 1/2 ipm, bar diameter three inches.

3.6 Components. The attachments shall consist of the following principal components.

3.6.1 Base and column assembly. The base and column assembly shall consist of, but not be limited to, a guide post(s) and vertical adjusting screw. The base plate shall be adaptable to the lathe and shall provide rigidity for the guidepost(s). A vertically adjustable lead-screw shall be furnished, mounted on anti-friction thrust bearings with take-up for end play, and engaged by anti-backlash nuts in the sliding casting. The lead-screw shall be of hardened alloy steel and shall have precision threads, which are uniform and with sufficient length of engagement to assure the adjusted positional accuracy specified in 3.5.2. The lead-screw shall be turned by a hand crank and shall be provided with an adjustable collar graduated in increments of 0.001 inch or 0.025 mm, which can be read continuously through the zero position.

3.6.2 Basic milling head. The basic milling head shall consist of a housing casting, spindle, gear drive, and motor support. The milling head shall be machined to house the spindle and gear drive assembly and to provide for motor attachment, guide post(s) slide, lead-screw, and positional clamp(s). The outer rim of the machined surface shall be graduated in increments of one degree for 30 degrees on each side of vertical zero. The spindle shall be hardened to Rockwell C45 to C52 and be precision ground, with the taper in the spindle nose conforming to ANSI B5.10 and Table I. The spindle and gear shafts shall be mounted on ball or roller bearings. All gears shall operate in an oil bath sealed to prevent entrance of foreign matter or leakage of oil when turned or rotated to any angle. The milling head shall provide for attachment of the drive motor, directly or by motor mount. The main spindle shall provide for positive drive for all side mill, end mill, shell end mill, angle mill, and fly cutter arbors.

3.6.3 Belts and pulleys. When applicable, V-belt drives shall be furnished in accordance with RMA IP 25. Pulleys shall be of cast iron or aluminum alloy and shall be balanced. A belt guard shall be provided for protection of the operator at all positions during metalworking or grinding operations.

3.6.4 Electrical equipment.

3.6.4.1 Motors. Motor(s) shall conform to CC-M-1807, as applicable, with Class B insulation. The motors specified in Table I shall be continuous duty, induction squirrel-cage types. The operating voltage, frequency, and phase shall

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be as specified (see Table I). Manual reset thermal overload protection shall be provided. A warning tag shall be attached to the power cord to identify the initial wiring of the motor (i.e., 200 V ac, 60 Hz).

3.6.4.2 Starting switch. The starting switch shall be manual and have a rated capacity in accordance with NEMA ICS 1. If toggle switches are furnished, they shall be provided with a guard to prevent accidental engaging of the switch.

3.6.4.3 Electrical components. Electrical components of the attachments, including starters, relays, switches, wiring, and their enclosures, shall conform to, and shall be located in accordance with, NEMA ICS 1 and NFPA 79, as applicable.

3.6.4.4 Power supply cord. Each attachment shall be furnished with a power supply cord properly wired into the attachment. The cord shall have a voltage and current carrying capacity equal to, or more than that required by, the motor(s) and NFPA 70. The power supply cord shall be at least ten feet long and terminate with a three prong grounded plug.

3.6.4.5 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.6.4.6 Electromagnetic interference control. When specified by the agency concerned, equipment shall require electromagnetic interference control in accordance with the requirements of MIL-STD-461 for Class IIB equipment. Test method CE03 and RE02 only shall be utilized.

3.7 Standard equipment. Each size attachment shall be furnished with (1) of each of the following pieces of standard equipment.

Size 1

Adapter, lathe compound slide, T-slot
Adapter, lathe carriage cross-slide
Arbor, angle mill, threaded RH, 3/8 24 UNF
Arbor, side mill, with 1/8 and 3/8 inch spacing collars and nuts for 7/8 inch bore cutters
Arbor, side mill, with 1/8 and 3/8 inch spacing collars and nuts for one inch bore cutters
Arbor, side mill, with 3 1/2 inch minimum mandrel, with 1/8, 3/16, 1/4, 1/2, 3/4, and 1 1/4 inch spacing collars and nuts for one inch bore cutters
Arbor, shell end mill, 1/2 inch
Collet or setscrew arbor for 3/8 inch straight shanks
Collet or setscrew arbor for 1/2 inch straight shanks
Arbor, flycutter, with 45 degree and 90 degree positions, sized to fit 1/4 inch minimum square cutting bit
Arbor, with 0 inch to 1/2 inch capacity geared chuck, with key
Belts, V, drive (one extra set)
T-bolts and nuts (for securing attachments to basic unit) (4 each)
Bolts (2)
Pulleys (2)

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Size 2

Adapter, lathe compound slide, T-slot
Adapter, lathe carriage cross-slide
Adapter for No. 2 ANSI machine taper shanks
Arbor, angle mill, threaded RH, 3/8 24 UNF
Arbor, side mill, with 1/8 and 3/8 inch spacing collars, standard keyway and nut for 7/8 inch bore cutters
Arbor, side mill, with 1/8 and 3/8 inch spacing collars, standard keyway and nut for one inch bore cutters
Arbor, side mill, with 3 1/2 inch minimum mandrel, with 1/8, 3/16, 1/4, 1/2, 3/4, and 1 1/4 inch spacing collars and nut for one inch bore cutters
Arbor, shell end mill, 3/4 inch
Arbor, for 3/8 inch diameter straight shanks, with setscrew
Arbor, for 1/2 inch diameter straight shanks, with setscrew
Arbor, flycutter, with 45 degree and 90 degree cutter positions, with 5/16 inch cutter bit
Arbor, with 0 inch to 1/2 inch capacity geared chuck, with key
Spiral cut end mill, 5/8 inch diameter, with 1/2 inch diameter shank
Spiral cut end mill, 3/4 inch diameter, with 1/2 inch diameter shank
Belts, V, drive (one extra set)
T-bolts and nuts (for securing attachments to basic unit) (4 each)
Bolts (2)
Pulleys (2)

3.7.1 Cutters. Each attachment shall be furnished with (1) set of high speed cutters as specified in table II. In addition to those listed, for size 1 units (1). Shell End Mill (size 1 1/2 X 1 1/8 X 1/2") and for size 2 units (1) Shell End Mill (size 2 X 1 3/8 X 3/4") shall be furnished. Tolerance of cutters shall be in accordance with ANSI B94.19.

TABLE II. Size of Various High Speed Steel Cutters[1]

Type of Cutter	Cutter Number[2]	Nominal Diameter of Cutter[3]	Diameter of Cutter	Width of face	Diameter of Shank	Diameter of Hole	Length of Cut	Overall Length
Woodruff	204	1/2	-	1/16	-	-	-	2 1/16
Keatseat	305	5/8	-	3/32	-	-	-	2 3/32
Milling	406	3/4	-	1/8	-	-	-	2 1/8
Cutters	507	7/8	-	5/32	-	-	-	2 5/32
(1/2 Inch	608	1	-	3/16	-	-	-	2 3/16
Straight Shank)								

[1]All dimensions are given in inches.
[2]The cutter numbers indicate nominal key dimensions or cutter sizes. The last two digits give the minimal diameter in eights of an inch and the digits preceeding the lat two given the nominal width in thirty-seconds of an inch. Thus cutter No. 204 indicates a size 2/32 X 4/8 inches or 1/16 X 1/2 inches in diameter.
[3]Furnished oversized to allow for sharpening.
[4]Threaded hole - Hand of rotation RH, hand of cutter RH.

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Plain Metal Slitting Saw	-	-	4	1/8	-	1	-	-
Staggered	-	-	2 1/2	1/2	-	7/8	-	-
Tool Side	-	-	3	1/4	-	1	-	-
Milling	-	-	3	5/16	-	1	-	-
Cutters	-	-	3	3/8	-	1	-	-
Two Flute	-	-	1/8	-	3/8	-	3/8	2 5/16
Single End	-	-	1/4	-	3/8	-	1/2	2 5/16
Mills for	-	-	3/8	-	3/8	-	9/16	2 5/16
Keyway Cutting	-	-	1/2	-	1/2	-	1	3
Single Angle Milling Cutter with Threaded Hole	-	-	1 1/4	7/16	-	3/8-24 UNF-2B RH[4]	-	-

3.7.2 Wrench set. Each attachment shall be furnished with (1) set of wrenches, which shall be capable of providing for the following functions:

Locking vertical adjustment of basic unit
Locking belt tension adjustment
Securing and removing V-belt sheaves and motor pulleys
Securing and removing all arbors and arbor nuts
Securing and releasing attachment fastening devices (see 3.8)
Securing and releasing belt-driven heads in shop cabinets

3.8 Optional equipment. When specified (see 6.2), the following optional equipment shall be furnished. Equipment shall be of proper size and capacity to fit the size of basic head specified. The basic attachment and all accessories shall be capable of performing within the specified accuracy without damaging any of the parts. The method of securing equipment to the basic attachment shall be accomplished without disturbing adjustments such as spindle bearing mountings.

3.8.1 Ninety degree universal milling head. The 90 degree universal milling head shall be capable of being securely mounted on the basic attachment. The head shall be adjustable for positioning the cutter at any point throughout 360 degrees and shall be self-indicating. The universal milling head shall permit the mounting of arbors and cutters specified herein and shall be capable of cutting the specified keyways in a 1020 steel round bar, 20 inches long, mounted between lathe centers. Cutting shall be at the specified range. The finish shall be not rougher than 125 microinches arithmetic average (aa). The cut shall extend full depth to the end of the bar with dimensions specified in table III.

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TABLE III. Size of Optional Equipment

Equipment	SIZE 1 Cut Dimensions Width (in.)	Depth (in.)	Length (in.)	Feed Rate (in./min)	Diam (in.)	SIZE 2 Cut Dimensions Width (in.)	Depth (in.)	Length (in.)	Feed Rate (in./min)	Diam (in.)
90 Deg Universal Milling Head	1/2	1/4	15	1	2[1]	3/4	3/8	15	1	3[1]
High Speed End Milling And Drilling Head	1/4	1/8	10	1	1[1]	3/8	3/16	10	1 1/2	1 1/2[1]
External Grinding Head	-	-	-	7	-	-	-	-	10	-
Internal Grinder[2]	-	3	-	-	1	-	4	-	-	1 1/4
Heavy-Duty Deep Hole Grinder[2]	-	8	-	-	2 1/4	-	10	-	-	2 1/2
Internal Keyseater and Slotter (Keyseat)	3/8	3/16	2 1/4	-	1 1/2	1/2	1/4	3	-	2
Feed Table (Keyway)	1/2	1/4	7	-	(3/4 X 3/4)[3]	3/4	3/8	7	-	1 X 1[3]

3.8.2 High-speed end milling and drilling head. The high-speed milling and drilling head shall be capable of being securely mounted on the basic milling head, shall mount the same arbors and tools specified for the basic attachment, and shall permit cutter axis adjustment no less than 30 degrees above to 30 degrees below horizontal. The head shall provide for mounting and operating tools at either side of the attachment without removing bearings. It shall provide a selection of spindle speeds arranged in approximate geometrical progression, and of proper range to provide for drilling and end milling operations. The head shall be capable of cutting the specified keyways, at the specified rate, in a 1020 steel round bar, 15 inches long, mounted between lathe centers. One keyway shall extend full depth to the end of the bar and the other keyway shall terminate 1/4 inch from the end of the bar as specified below. The cut surfaces shall be not rougher than 125 microinches aa with dimensions specified in table III.

[1]Bar diameter.
[2]Allowance on rough hole size +1/8 inch, -0 inch.
[3]Minimum bar size.

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3.8.3 External grinding head. The external grinding head shall be a precision machine capable of cylindrical, tool, and thread grinding. The head shall be capable of being mounted on the basic milling head, shall permit grinder axis adjustment not less than 30 degrees above to 30 degrees below horizontal, and shall be driven by the basic milling head motor. Provision shall be made to permit grinding from either side of the attachment without disturbing the spindle-bearing mounting. A minimum of six spindle speeds shall be provided. Grit, bond, diameter, and the speed of the grinding wheels furnished shall be suitable for efficient grinding at the full diameter of the wheels, and after the wheels have been dressed to 2/3 of their original diameter. Grinding-wheel speeds shall in no case exceed the requirements of ANSI B7.1. The following grinding wheels shall be furnished with each head:

Two straight wheels
One flare cup wheel
One dish wheel

A grinding wheel guard conforming to the requirements of ANSI B7.1 shall be provided. The grinder shall be capable of removing not less than 0.020 inch from the diameter of a 1 inch (+1/4 inch, -0 inch) shaft of 1045 steel, heat treated to Rockwell C40 to C50. Length of ground surface shall be not less than 10 inches. The specified diameter reduction shall be acccomplished in not more than 30 minutes and the resulting surface finish shall be not rougher than 32 microinches aa. Feed rate shall be as specified in table III.

3.8.4 Internal grinder. The internal grinding head shall be similar in construction to the external grinding head and subject to the same requirements, except that the spindle shall be extended for grinding bores of the diameter and depth specified by table III in 1045 steel, heat treated to Rockwell C40 to C50. The grinder shall be capable of removing 0.020 inch from the bore diameter in not more than 30 minutes. Surface finish shall not be rougher than 16 microinches aa. A minimum of (6) spindle speeds shall be provided. Two straight wheels shall be furnished with each grinder.

3.8.5 Heavy-duty, deep-hole grinder. The heavy-duty, deep-hole grinder shall be capable of being mounted on the basic milling head and shall conform in general characteristics of design to other grinding heads specified herein. The grinder shall be capable of grinding bores of the diameter and depth specified by table III in 1045 steel, heat treated to Rockwell C40 to C50. The grinder shall be capable of removing 0.020 inch from the bore diameter in not more than 30 minutes. The surface finish shall be not rougher than 16 microinches aa. A minimum of (6) spindle speeds shall be provided. A grinding wheel shall be furnished that shall allow this requirement to be accomplished.

3.8.6 Dividing head. The dividing head shall be adaptable to fit 1 3/8 inch diameter lathe spindle bores. The head shall be furnished with one machined adapter for a 1 3/8 inch bore spindle and three unfinished adapters to cover a range of 1 1/8 inches to 3 1/2 inches. The dividing head shall be capable of indexing all divisions from one to 50 inclusive, and divisions 55 to 120 by multiples of five. The dividing head shall index by degrees (360 equal divisions), 1/2 degree (720 equal divisions), and 1/3 degree (1080 equal divisions). Positioning error shall be not greater than 1/2 minute of one degree (30 seconds).

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3.8.7 Internal keyseater and slotter. The internal keyseater shall be capable of cutting a standard internal (as listed below) keyway in the bore size, for which it is standard, for a length up to 1 1/2 times the bore diameter (as listed in ANSI B17.1). Sizes 1 and 2 shall have a stroke adjustable from 0 to at least 4 inches. The keyseater and slotter shall be capable of being securely mounted on the basic attachment and of being locked in position to provide strokes of any angle in relation to the base mounting surface. It shall include the necessary cutters and tool bits to cut each of the internal keyways specified below for both Size 1 and Size 2.

1/8 by 1/16 inch
3/16 by 3/32 inch
1/4 by 1/8 inch
5/16 by 5/32 inch
3/8 by 3/16 inch
1/2 by 1/4 inch

The keyseater and slotter shall be capable of cutting the keyseats specified in table III, in the specified bores of 1020 steel, in not more than five minutes. The surface finish shall be not rougher than 125 microinches aa.

3.8.8 Feed table. The feed table shall be capable of being mounted to the basic attachment to create a portable machining unit. It shall consist of a base, a lower carriage that shall travel longitudinally in relation to the base, an upper carriage that shall travel across the base, and the necessary screws, screw blocks, anti-backlash nuts, and gibs to equip it to meet the accuracy and performance specified herein. The base shall be capable of being rigidly bolted or clamped to a supporting surface without interfering with the movement of either carriage. The lower carriage shall provide travel of not less than 3 inches at 90 degrees to the direction of travel of the lower carriage. A means for locking at any point in either direction of travel shall be provided. Locking either carriage shall not deflect the basic attachment more than 0.001 inch. Feed shall be provided with a graduated collar or dial similar in form and marking to that specified for vertical travel of the basic unit. Each collar shall be capable of being set at 0 at any point of the carriage travel. Travel, as indicated by either collar, shall be accurate within 0.005 inch per inch of travel. The basic head, when mounted on the feed table, shall permit cutting the keyways specified by table III in a 1020 steel square bar in not more than 5 minutes for size 1 or 6 minutes for size 2. Feed shall be manually accomplished. The total height from the mounting surface to the top surface shall not exceed 3 3/8 inches.

3.8.9 Grinding gage. A grinding gage (tooth stop rest) for milling cutters shall be universally adjustable and sufficiently rigid to provide accuracy in grinding cutters.

3.8.10 Diamond dresser. The diamond-wheel dresser shall consist of a frame which can be clamped in a suitable position for dressing grinding wheels specified herein, and a properly mounted industrial diamond. The diamond shall be free from faults that may affect the serviceability of the dresser. Diamonds shall be natural octahedron, with not less than (6) cutting edges. Diamonds shall be at least 1/4 carat if the maximum size of the wheels furnished is 4 inches or less in diameter, at least 3/10 carat if the maximum size wheels furnished is 6 inches, and not less than 1/2 carat for 7 inch wheels.

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3.8.11 Shop cabinet. A shop cabinet shall be provided of sufficient size, with brackets, clamps, etc., to store the basic attachment and all specified and optional equipment, spare parts, and wrenches in an orderly manner, and such that the various stored parts shall not be displaced by rough handling due to shipping. The cabinet shall be designed to provide easy access to its contents. It shall be adequate to protect all contents from air-blown metal chips, coolant, dirt, grease, and other contaminants encountered during machine shop operations.

3.9 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.10 Product identification. The attachments shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.11 Workmanship. Standards of workmanship shall assure that the attachments shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).

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(b) Motor tests (see 4.6.1.1).
(c) Milling test (see 4.6.1.2).
(d) Woodruff keyseat cutting test, with basic attachment (see 4.6.1.2.1)
(e) Packaging inspection (see 4.7).

4.4.1 Sampling. Unless otherwise specified (see 6.2), sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination (4.5)	100 percent	No defects
Motor tests (4.6.1.1)	S-4	1.0
Milling test (4.6.1.2)	S-4	1.0
Woodruff keyseat cutting test (4.6.1.2.1)	S-4	1.0
Packaging inspection (4.7)	S-4	4.0

4.5 Product Examination. Visually and manually examine the attachment to determine conformance with the requirements of 3.2 through 3.5.2 and 3.6 through 3.11. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the attachment has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection. A certificate of conformance may be accepted in lieu of the visual or manual requirements of 3.2.4, 3.2.5.1, 3.4.1, 3.4.2 and 3.4.5.

4.5.1 Evidence of compliance with OSHA standards. Evidence of compliance with applicable safety and health requirements shall be reviewed (see 3.2.5.1 and 6.5).

4.6 Tests.

4.6.1 Performance tests. The attachment and the standard and optional equipment (see 3.7 and 3.8) shall be tested as follows for compliance with the requirements of 3.5.

4.6.1.1 Motor tests. The contractor shall certify that all motors have been tested and comply with the applicable test requirements of CC-M-1807 (see 3.6.4.1).

4.6.1.2 Milling tests. The basic attachment shall be set up on a suitable lathe (see 3.5.3). Keyways of specified size and lengths shall be cut in one continuous operation, at the specified rate, in 1020 steel of a diameter appropriate for use with a plain parallel key in accordance with ANSI B17.1 except as specified herein. Cuts shall be straight and true and the finish shall be within 125 microinches aa. A follower rest, or other method of bracing the stock, may be used. Cuts shall not vary from a straight line more than 0.002 inch per foot. Set-up and milling shall be in accordance with information contained in the manufacturer's instructions.

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4.6.1.2.1 Woodruff keyseat cutting with basic attachment. Two Woodruff keyseats of appropriate size shall be cut to determine compliance with 3.5.3.1.

4.6.1.2.2 Keyway cutting with basic attachment. A keyway of appropriate size shall be cut to determine compliance with 3.5.3.2.

4.6.1.2.3 Keyway cutting with 90 degree universal milling head. The universal milling head shall make the specified cuts, with the spindle in a vertical position to determine compliance with 3.8.1.

4.6.1.2.4 Keyway cutting with the high-speed end milling and drilling head.. The end milling head shall make two cuts to determine compliance with the requirements of 3.8.2.

4.6.1.3 Grinder tests. The basic attachment, with appropriate grinding head, shall be set up on a suitable lathe. Set up and method of obtaining power feed rates shall be in accordance with information contained in the manufacturer's instructions.

4.6.1.3.1 Performance test of external grinder. The work piece shall be 15 inches long and rough turned to present a test surface not less than ten inches long. The test surface diameter shall be reduced 0.020 inch to determine compliance with 3.8.3.

4.6.1.3.2 Performance test of internal grinder. The internal grinder shall be tested to determine compliance with the requirements of 3.8.4.

4.6.1.3.3 Performance test of deep-hole grinder. The deep-hole grinder shall be tested to determine compliance with the requirements of 3.8.5.

4.6.1.4 Performance of dividing head. The dividing head shall be properly mounted on a lathe, and test cuts shall be made in the periphery of a six inch diameter workpiece held by the lathe chuck. The cuts shall be made with a side-milling cutter and shall be 1/4 inch wide and 1/8 inch deep. Cuts shall be spaced at 120 degrees, 90 degrees, 60 degrees, 40 degrees, and 20 degrees from the first cut. The total error in location of each cut shall not be more than one-half minute of one degree when measured from the base cut (see 3.8.6).

4.6.1.5 Performance test of internal keyseater and slotter. The keyseater and slotter shall be tested to determine compliance with the keyseat cutting requirements of 3.8.7.

4.6.1.6 Performance test of feed table. The basic attachment, with feed table, shall be securely attached to a suitable surface fitted with a vise or other means for positioning and holding the appropriate work piece. A keyway shall be cut to determine compliance with the requirements of 3.8.8.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Military preservation, packing, and marking. When specified (see 6.2), requirements for Level A preservation, Levels A or B packing, and military

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marking shall be in accordance with MIL-M-18058 except that the basic attachment shall be prepared for shipment as specified in MIL-E-16298.

5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

5.3 Marking for civil agencies. When specified (see 6.2), marking for shipment for civil agencies shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The attachment covered by this specification is intended for use in milling, drilling, boring, gear cutting, thread milling, keyseating, slotting, and external and internal grinding operations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2.1).
(c) First article when required (see 3.1).
(d) Measuring and indicating device graduation(s) required (see 3.2.2).
(e) Optional speed range and motors when required (see 3.5 and Table I).
(f) Voltage, current, frequency, and phase required (see Table I and 3.6.4.1).
(g) Optional equipment required (see 3.8 thru 3.8.11).
(h) Painting and finishing if different (see 3.9).
(i) Product identification if different (see 3.10).
(j) First article inspection if different (see 4.3).
(k) Quality conformance inspection if different (see 4.4).
(l) Sampling when required (see 4.4.1).
(m) Level of preservation, packing, and marking if different (see 5.1 and 5.2).
(n) Civil agency shipment marking when required (see 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract (see 6.3.1).

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, and price. Repair part identification should also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items should be so identified as shown in MIL-M-7298,

6.4 Standard lubricants. The attachment manufacturer should specify and use to the maximum extent possible those lubricants (see 3.4.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

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6.5 OSHA standards. Products which meet all requirements of this specification comply with the OSHA standards published prior to the date of this specification and identified in Section 3 (see 3.2.5.1).

MILITARY INTERESTS:

Custodians -

Army - AL
Navy - SH
Air Force - 99

Review Activities

Army - ME
Navy -
Air Force - 84
DLA - GS

User Activities

Army - AV, CE
Navy - MC, YD
Air Force -

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

Army - AL

DoD Project 3460-0118

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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